

Electronic chip that mimics human brain

Scientists have developed an electronic chip that can mimic the human brain and could lead to a new generation of powerful, energy-efficient electronics.

Researchers at University of Twente in The Netherlands have demonstrated working electronic circuits, which can be compared to their conventional counterparts in terms of size, but are much closer to natural networks like the human brain.

Computers have become more and more powerful by integrating ever smaller components on silicon chips, but it is becoming increasingly difficult and expensive to continue this miniaturisation. Current transistors consist of only a handful of atoms.

Researchers have used networks of gold nanoparticles for executing essential computational tasks. They have moved away from designed circuits and by using design-less systems, have avoided costly design mistakes.

Computational power of their networks is enabled by applying artificial evolution, which takes less than an hour. By applying electrical signals, the same network can be configured into 16 different logical gates.

Smartglove turns sign language into text

Hadeel Ayoub, designer and student at Goldsmiths, University of London, UK, has developed a smart-glove called SignLanguageGlove that features sensors to convert hand and finger movements into text and spoken dialogue.

The first experimental version of the glove was equipped with five flex sensors to monitor the bends and curves of each finger and an accelerometer to detect the orientation of the glove. It took signs and turned these into visual letters on a screen. Ayoub then followed up with an improved model that was faster and more robust, featuring smaller, more discreet hardware and text that scrolled on a screen.

The latest model features a text-to-speech chip with the hardware sewn into the lining of the glove. He is now integrating a language-translation function into the system.



Schematic of a nano-particle network (about 200 nanometres in diameter). By applying electrical signals at the electrodes (yellow) and using artificial evolution, this disordered network can be configured into useful electronic circuits (Image courtesy: University of Twente)

Create buttons with a gel touchscreen

Researchers from Technische Universität Berlin in Germany have built a prototype of a touchscreen with a layer of gel atop it that can change from soft to stiff when heat is applied, making it possible to create temporary buttons in all kinds of shapes that need not be defined in advance, which users can feel and use to interact with the display.

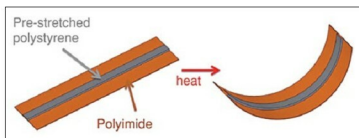
This technology could make it easier to use a range of electronics, from in-car displays to smartphones and wearable gadgets, and to do things like receive alerts or input information without needing to glance at the devices.

Using the prototype, a 17.8cm (7-inch) touchscreen called GelTouch, researchers stiffened the gel into three basic shapes to form a grid of buttons, a slider and a one-finger joystick button. They tried these out with several different sample applications such as using buttons to dial a phone number without looking at the display, the slider to scroll through photos and the joystick to play a simple game. A heat-responsive hydrogel was used atop the display, which is transparent and jelly-like until it is heated above 32.2°C (90°F).

In the prototype, it takes about two seconds to harden once heat is applied and about the same amount of time to soften again.

Self-wrapping 3D electronics are in

A team of researchers at Stanford University, USA, has developed shape-controlled, self-wrapping 3D single-walled carbon nanotube (SWNT) electronics us-



Upon heating, the bilayer structure changes shape (Image courtesy: www.materialsvIEWS.com/custom-fit-self-wrapping-3d-electronics)

ing a bi-layer structure of shape memory polymers and polyimide. For the first time, a shape memory polymer is used in electronic devices, with performance demonstrated in the form of transistors, gas and temperature sensors and memory devices.

Researchers achieved local shape control with electrically-activated pre-patterned heaters, which allow the curvature to be tuned in various directions at different times. They fabricated a foldable lotus flower that folds up on increasing the heating temperature from 110°C to 140°C. Further heating beyond 190°C caused the flower to unfold again, as the polystyrene softened.